

Department of Cancer Research and Molecular Medicine

Examination paper for MOL3100 - Introduction to Molecular Medicine with project MOL3000 - Introduction to Molecular Medicine

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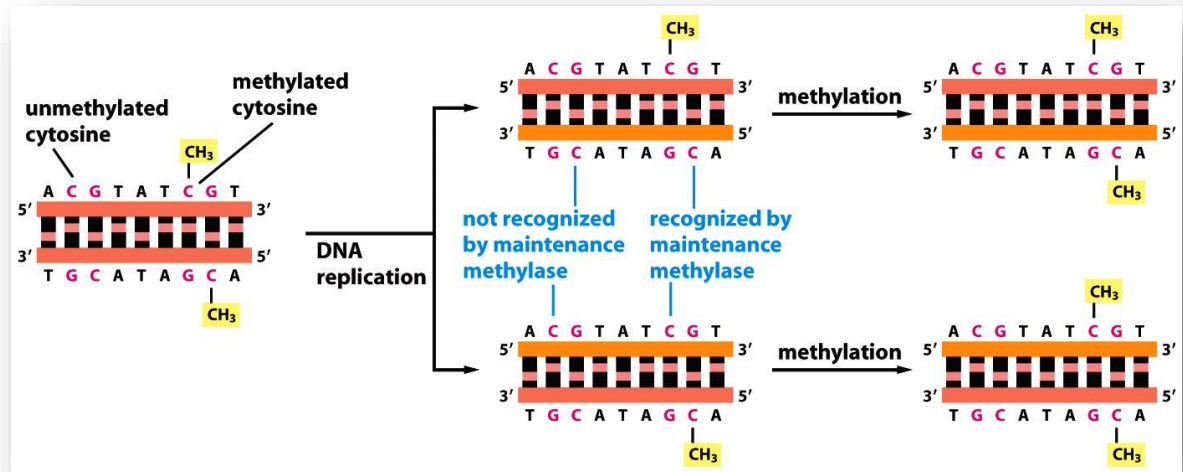
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Question 1 (6p)

Epigenetic mechanisms drive differentiation and development and are involved in complex biological processes such as learning and memory. The best studied epigenetic mark is DNA methylation. DNA methylation occurs mainly on cytosine when positioned before guanine (5'-CpG-3' context).

A) Describe the process of DNA methylation by DNA methyltransferase DNMT1. (2p)
DNMT1 copies the methylation pattern from the DNA template (recognizes 5mC on the parental strand) to the new strand during replication (maintenance methylation).



B) Epigenetic modifications are dynamically regulated as part of a biological response. Describe the molecular process of active demethylation of 5-methylcytosine (5mC). (2p)

Enzymatic process – TET (ten-eleven translocase) mediated oxidation of the methyl-group of 5mC to 5hmC (hydroxymethyl-C), 5fC (formyl-C) and 5caC (carboxy-C).

1) These oxidized forms are not recognized by DNMT1 and will result in unmethylated C on the nascent strand after replication.

2) 5fC and 5caC are substrates for the DNA glycosylase TDG (thymine DNA glycosylase), which cuts them out of DNA. The resulting abasic sites are repaired by base-excision repair (BER) (AP-endonuclease, polymerase, ligase)

C) Discuss briefly how aberrant DNA methylation may be involved in disease. (2p)

Aberrant DNA methylation may result in disease as a consequence of both transcriptional activation (e.g. oncogene activation) and transcriptional repression e.g. inactivation of tumor suppressors). Hypomethylation of transposons may result in activation and genomic instability.

Question 2 (4p)

Use of stem cells is important in medical research and therapy.

A) Name three examples of therapeutical applications of stem cells.(2p)

e.g.

Hematopoietic stem cell transplant

Stem cells in skin transplant approaches

Use in treatment of diabetes

Potential in Parkinson disease

Induced pluripotent stem cells are obtained by reprogramming differentiated adult cells, e.g. fibroblast from a skin biopsy.

B) Name four factors used by Yamanaka and coworkers to generate induced pluripotent stem cells and indicate their common cellular function. (2p)

c-Myc, Klf4, Sox2, Oct4

These function as transcription factors.

Question 3 (6p)

More than 4000 monogenic disorders have been described. They are most commonly transmitted by autosomal or X-linked modes.

- A. What will be typical observations suggesting an X-linked recessive inheritance mood of a condition? (1p)

Preferred answer:

Incidence much higher among men. Heterozygous women usually unaffected (healthy carriers). If the father is sick, all daughters will be carriers. "Male-to-male" transfer of X-linked conditions is not possible since a son inherits his X-chromosome from his mother. Families with an X-linked recessive disorder often have affected males, but rarely affected females, in each generation.

- B. For an X-linked dominant non-lethal disease, the prevalence among girls is higher than among boys. What is the most likely explanation for this observation? (1p)

Preferred answer:

Males can only get an X chromosome from their mother whilst females get an X chromosome from both parents. As a result, females tend to show higher prevalence of X-linked dominant disorders because they have more of a chance to inherit a faulty X chromosome.

- C. Fragile X syndrome affects about 1 in 5000 boys and is caused by tri-nucleotide (CGG) expansion in the 5' untranslated region (UTR) of *FMR1* gene.
Explain the molecular mechanism that leads to silencing of *FMR1* expression in fragile X. (2p)

Preferred answer: The length of unstable CGG repeats increases for each generation of maternal transfer. A full mutation develops if the number of repeats extend beyond 200 copies. Methylation at CpG dinucleotides in the 5' UTR region leads to the recruitment of methyl-binding protein complexes able to modify the chromatin structure into a transcriptionally inactive state, which spreads further into the *FMR1* gene and turns it off.

- D. Fragile X was previously classified as an X-linked dominant condition but shows a more complex inheritance pattern due to variable expressivity, reduced penetrance and genetic anticipation.
Explain what is meant by genetic anticipation. (1p)

Preferred answer:

Genetic anticipation: earlier onset and worsening of symptoms in subsequent generations.

Some mitochondrial diseases (mitochondriopathies) show a clear maternal inheritance pattern, but with huge variation in the severity of symptoms among offspring that is independent of gender.

E. What is the rationale for this huge variation in symptoms? (1p)

Preferred answer: Mitochondriopathies can be caused by mutations in the mitochondrial genome (mtDNA) (~15%), which is inherited from mother (egg cell).

Each cell contains many mitochondria (usually > 1000). Each mitochondrion contains several mtDNA molecules (2-10). These are usually identical (homoplasmy) but may also be a mixture of mutant and wild type versions (heteroplasmy). The number of normal and mutated mitochondria may vary substantially between different egg cells. Children may therefore be differentially affected. The mother may not be affected

Question 4 (4p)

One of the first genetic aberration discovered in cancer was the so-called Philadelphia chromosome in chronic myelogenous leukemia (CML). This chromosome results from a balanced translocation between chromosomes 9 and 22, leading to production of a fusion protein produced from parts of the *BCR* gene from chromosome 22 and the *ABL* gene from chromosome 9.

- A) Describe briefly why CML is regarded as a genetically unique disease among cancers (2p).

Answer: In CML the abovementioned translocation is required and enough to cause the disease while development of other cancers requires a number of genetic changes in both oncogenes and tumor suppressor-genes. In many cases these mutations may be in different genes, even in cancers with the same diagnosis and arising from identical tissues. In CML, aberrant expression of the ABL-kinase activity is sufficient to drive cancer development, and inhibition of the ABL-tyrosine kinase cures most of the patients.

- A) One method to detect the Philadelphia chromosome is by FISH (Fluorescent In situ hybridization). By which other method can the *BCR-ABL* translocation relatively easily be detected in a blood sample? (1p)

Answer: PCR (or RT-PCR) on extracted DNA (or RNA) from blood with primers flanking the translocation point.

Hva med FISH? Bør jo også gi rett svar

- B) CML is usually treated with the tyrosine-kinase inhibitor Imatinib (Gleevec). Why do some patients become refractory to this targeted treatment? (1p)

Answer: Imatinib eradicates leukemic cells due to inhibition of BCR-ABL by Imatinib binding, but a few cells may have (or acquire during treatment) mutations in the Imatinib-binding pocket of ABL. Cells with such mutations may survive the treatment, and lead to relapse of the CML.

Question 5 (4p)

The bacterium *Escherichia coli* is one of the most common causes of infections in humans.

A) Describe which types of infection this bacterium may cause.

Sensor guide:

1) Intestinal infections- several different pathotypes of *E. coli* cause diarrhea. The different pathotypes are defined by the molecular mechanism by which they cause disease. 2) Extraintestinal infection: Urinary tract infection, sepsis/bloodstream infection and meningitis.

B) Why is antimicrobial susceptibility testing done primarily by phenotypic methods rather than PCR-based methods in medical microbiology laboratories?

Sensor guide: Antimicrobial resistance is often due to one or more mutations, which may be difficult to detect by PCR, and new mechanisms may occur that will not be detected by current molecular methods, while phenotypic methods will detect reduced susceptibility independent of mechanisms.

Question 6 (6p)

In 1920 Frederick Banting and Charles Best demonstrated that they could reverse induced diabetes in dogs by giving them an extract from the pancreatic islets of Langerhans of healthy dogs. About two years later, Leonard Thompson (a 14-year boy) became the first person with diabetes to receive insulin injection. He lived another 13 years before dying of pneumonia.

A) Describe the islets of Langerhans, particularly regarding endocrine cell types (2p).

Answer:

- alpha-cells produce glucagon
- beta-cells produce insulin
- delta-cells produce somatostatin

B) Which type of diabetes did Leonard Thompson have? Describe short the molecular pathogeneses of type 1 and type 2 diabetes (2p).

Answer

- He was T1DM
- T1DM pathogenesis includes genetic predisposition, auto-immune reaction, beta-cell damage and clinical complications
- T2DM pathogenesis includes impaired glucose tolerance, insulin resistance, and (secondary?) beta-cell dysfunction

C) Insulin is used as a life-supporting molecular medicine (e.g. Leonard Thompson). Describe possible curative strategies for diabetes based on your knowledge (2p).

Answer (one of the following):

- Pancreatic transplantation
- Islet cell transplantation
- Stem cell transplantation
- Beta-cell regeneration
- Artificial pancreas

MCQ (20 p)

Only one letter (A,B,C or D) for each question (the best alternative). Make a table and hand in your answers together with the essay questions.

1.

DNA methylation occurs on cytosines positioned before guanine (CpG context).	
In what regions in the genome are the CpG sites often unmethylated?	
A	exons
B	introns
C	Transposable elements
D	CpG islands

2.

A healthy 37-year old women with normal cholesterol levels has a 40-year old brother suffering from familial hypercholesterolemia (FH). Both their parents are known to be heterozygous for the same severe FH mutation.	
What is the chance that the women will also be heterozygous for the FH mutation?	
A	~0
B	1/4
C	2/3
D	1/2

3.

Aneuploidies involve the net loss or gain of chromosomes.	
Which of the following aneuploidies would be expected not to be viable at birth?	
A	47, XX + 13
B	45, XY – 12
C	45, X
D	47, XY + 18

4.

Twin studies, comparing monozygotic (MZ) and dizygotic (DZ) twin pairs, are often used to study the relative contribution of genes and environment to complex diseases.	
Whish statement about concordance rates in twin-studies is correct?	
A	Greater concordance rates in DZ twins than in MZ twins is considered evidence that a disease has a genetic component (i.e. is influenced by genetics)
B	Less than 100% concordance in MZ twins is considered evidence that a disease is also influenced by the environment
C	Similar concordance rates in MZ and DZ twins is considered evidence that genetics play a major role in the development of the disease
D	Greater concordance rates in MZ twins than in DZ twins is considered evidence that a disease is also influenced by the environment

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5.

In medical microbiology we use the term opportunistic microbe.	
What does this term mean?	
A	A microbe that causes infection in immunocompromised persons
B	A microbe that colonizes but does not cause symptomatic infection in humans
C	A microbe that causes slowly evolving chronic infection in humans
D	A microbe that very rarely causes infection in humans

6

The <i>RAS</i> oncogene is activated in a large fraction of human cancers by recurrent point mutations changing amino the acid composition of the RAS protein.	
What consequences do these amino acid changes have?	
A	The mutations facilitate stronger GTP binding.
B	The mutations facilitate stronger GDP binding
C	The mutations facilitate stronger ATP binding.
D	The mutations facilitate stronger ADP binding

7.

Cardiac repair after myocardial infarction is characterized by an inflammatory phase initiated by danger associated molecular pattern (DAMP) releasing cells	
Which of the following statement is most true?	
A	DAMPs are released by infiltrating leukocytes
B	DAMPs are released by cardiac macrophages
C	DAMPs are released by infiltrating neutrophils
D	DAMPs are released by damaged cardiomyocytes

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Endothelial cell dysfunction is a central part in the process of atherosclerosis	
Which factors contribute to impaired endothelial cell function?	
A	increased level of nitric oxide and increased level of oxygen radicals
B	decreased level of nitric oxide and decreased level of oxygen radicals
C	increased level of nitric oxide and decreased level of oxygen radicals
D	decreased levels of nitric oxide and increased level of oxygen radicals

9.

Non-coding RNA is involved in many biological processes.	
Which of the following non-coding RNA species is involved in the silencing of transposons?	
A	piRNA
B	snRNA
C	snoRNA
D	miRNA

10.

Aptamers are useful in therapeutics and diagnostics.	
How do aptamers work?	
A	By downregulating protein expression at a post-transcriptional level
B	By silencing transposons
C	By binding specific ligands with high affinity through steric interactions
D	By cleaving transcripts

11.

lncRNAs are classified according to where they are encoded in the genome relative to protein-coding genes.	
What would we call a lncRNA that is encoded between two protein-coding genes?	
A	Intronic lncRNA
B	Antisense lncRNA
C	Intergenic lncRNA (lincRNA)
D	piRNA

12.

Epigenetic regulation includes modification of the N-terminal histone tails	
Histone methylation occurs on amino acid residues:	
A	Serine and Threonine
B	Lysine and Arginine
C	Tyrosine and Lysine
D	Tyrosine and Arginine

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In the Computer Lab exercise the PDB (Protein Data Bank) database was used for searching.	
Why was that particular database used?	
A	The PDB database is specific to proteins, rather than DNA.
B	We wanted to focus on sequences with known 3D structure.
C	There is no particular reason, except that it is part of the PSI-Blast suite of databases.
D	It is one of the most extensive databases of protein sequences available, which ensures a maximum number of hits.

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A typical pairwise alignment as shown in the output of PSI-Blast consists of 3 lines. Two of the lines represent the query sequence and a specific database hit, whereas the third line represents the similarity between these two sequences.	
In the line representing similarity, what does “+” indicate?	
A	The “+” is indicating that the pair of residues are found in similar 3D structures.
B	The “+” represents amino acids with a positive net charge.
C	The “+” is just a ruler used per 10 residues of alignment.
D	The “+” represents a pair of amino acids with a positive similarity score (also known as profile score) in the score matrix.

15.

When comparing sequences by sequence alignment we need to estimate the significance of the similarity. The E-value is often used for this.	
What does the E-value actually represent?	
A	Expected number of aligned sequence positions in the optimal alignment.
B	Expected number of database sequences that give at least the same score value if the database is searched with random query sequences.
C	Expected number of identical sequences in the database.
D	Expected number of database sequences that give at least the same score value if the database is searched again with the same query sequence.

16.

Mutations are often involved in disease, but also required for evolution	
A mutation is defined as:	
A	Any type of damage to DNA
B	A change in that DNA sequence that causes disease
C	A nucleotide sequence in DNA that deviates from the normal
D	A nucleotide sequence in a gene that deviates from the normal

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17.

Hemoglobin A1C measures the percentage of hemoglobin that is glycated and determines average blood glucose during the two to three months prior to testing.	
Blood sugar is well controlled when HbA1C is:	
A	Below 7%
B	Between 12%-15%
C	Less than 180 mg/dL
D	Between 90 and 130 mg/dL

18.

Persons with elevated glucose levels that do not yet meet the criteria for diabetes are considered to have prediabetes.	
Prediabetes is associated with all of the following except:	
A	Increased risk of developing type 2 diabetes
B	Impaired glucose tolerance
C	Increased risk of heart disease and stroke
D	Increased risk of developing type 1 diabetes

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Type 1 diabetes is a primary failure of pancreatic beta cells to produce insulin. It primarily affects children and young adults.	
The risk factors for type 1 diabetes include all the following except:	
A	Diet
B	Genetic
C	Autoimmune
D	Environmental

Deamination of cytosine may lead to mutation if not repaired by the enzyme uracil-DNA glycosylase before the next round of replication	
Which mutation?	
A	C > A
B	C > T
C	C > G
D	C > U

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